

ECON 0100 | Exercise C1 | Externalities

Hogsmeade has seen a recent proliferation of candy shops selling smoked toffees. The Demand (marginal benefit) curve and Supply (marginal cost) curve for smoked toffee can be represented using the following relationships.

$$D : P = 100 - Q_d \qquad S : P = 10 + Q_s$$

The proliferation of smoked toffee makers has meant the town saw a subsequent rise in *smoke*. Wood smoke can cause lung issues and carries a cost of 10 Galleons to the health of Hogsmeade residents.

Q1 | One Toffee Exchange

First Exchange: A buyer values one smoked toffee at 60 Galleons. It costs the seller 55 Galleons to make.

- a) Will the buyer and seller make this exchange? _____
- b) Total Surplus to the buyer and seller: _____
- c) Marginal Social Cost: _____
- d) Total Surplus, including the residents: _____
- e) Deadweight Loss from this exchange: _____

Another buyer values one smoked toffee at 80 Galleons. It costs the seller 40 Galleons to make.

- f) Will the buyer and seller make this exchange? _____
- g) Total Surplus to the buyer and seller: _____
- h) Marginal Social Cost: _____
- i) Total Surplus, including the residents: _____
- j) Deadweight Loss from removing this exchange: _____

Q2 | Market Equilibrium

Use a graph and algebra to find equilibrium price, quantity, and DWL. Be sure to show your work.

Marginal Social Cost: _____

Market Equilibrium Price: _____ Market Equilibrium Quantity: _____

Socially Efficient Quantity: _____ Deadweight Loss: _____

Q3 | Yard Flowers

As a small residential town, Hogsmeade has both 1) many shops, and 2) many homes near the town center. Residential homeowners often plant flowers in their front yards, contributing to the positive atmosphere in town. There are numerous garden stores selling supplies for planting and growing many varieties. The supply and demand relationships for a dozen (generic) flowers is given by the following equations, with prices in Galleons.

$$D : P = 100 - \frac{1}{2}Q_d \qquad S : P = \frac{1}{2}Q_s$$

Recently, a study found a positive impact on the wellbeing of the community from the flowers planted in the front yards of their neighbors. The impact was estimated to carry a value of 10 Galleons per dozen flowers planted. Use a graph and algebra to calculate the size of the market failure by identifying and calculating the deadweight loss.

Market Equilibrium Quantity: _____ Socially Efficient Quantity: _____

Deadweight Loss: _____